UNIVERSITY OF CAMBRIDGE

NATURAL SCIENCE TRIPOS Part 1B

PHA/WP

PHARMACOLOGY WRITTEN PRACTICAL PAPER

Paper available from 1 June 2020 (12:00) – 29 June 2020 (23.59)

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*Answer **all three** questions.*

You have 3 hours (plus any pre-agreed individual adjustment) to answer this paper.

Please treat this as a closed book exam and write your answers within the time-period. Downloading and uploading times should not be included in the allocated exam time.

Assignments should be hand-written (and legible).

You are not required to plot any graphs.

You are required/expected to show the mathematical process (e.g. write the equations) by which you reached your answers.

You will not need to provide figures (although if you wish to scribble things within your answers, you may do so)

No candidate will be penalised for spelling or grammatical errors.

Candidates are permitted to use an approved calculator.

***IMPORTANT: Each assignment must be saved as a single file**, i.e. after the assessment is complete assemble your assignments into three separate files for upload.*

After completing the assessment follow the guidance documents for information on formatting each of your assignments and for upload via the Moodle Assessment site.

*After completing the assessment please download, complete and upload the **coversheet** (Coversheet_Pharmacology_Written_Practical_Paper) stating the questions you answered.*

Question 1

During their novel bronchodilator discovery campaign, CB2 pharmaceuticals designed and developed a preclinical candidate - Millerein which chemically exists as mixture of two optical isomers namely (+)-Millerein and (-)-Millerein (Figure 1). Experiments were carried out by the CB2 drug discovery team to evaluate the properties of these two isomers against guinea-pig tracheal smooth muscle contraction evoked by cholinergic stimulation. The extent of contraction of spiral strips of guinea-pig trachea in response to various concentrations of carbachol in absence and presence of 0.1µM of (+) and (-) Millerein are given in Table 1 and shown in Figure 2.

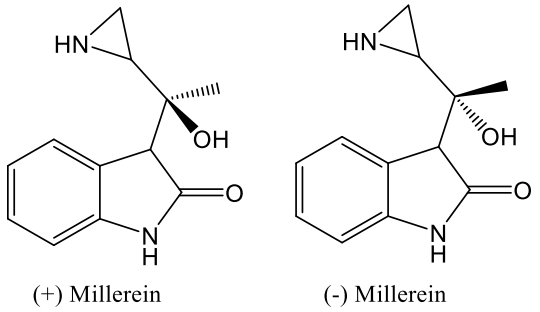

Figure 1. 2D chemical structures of Millerein isomers

Table 1. *Contractions of guinea-pig tracheal strips to carbachol*

Carbachol concentration (µM)	Height of contraction (mm on chart recorder)		
	Control	with (-)-Millerein	with (+)-Millerein
0.1	4	1	1.3
0.5	24	11	13
1	45	23	28
2	75	45	59
4	105	76	87
10	131	112	120
20	141	130	126
40	146	139	143
100	150	151	148

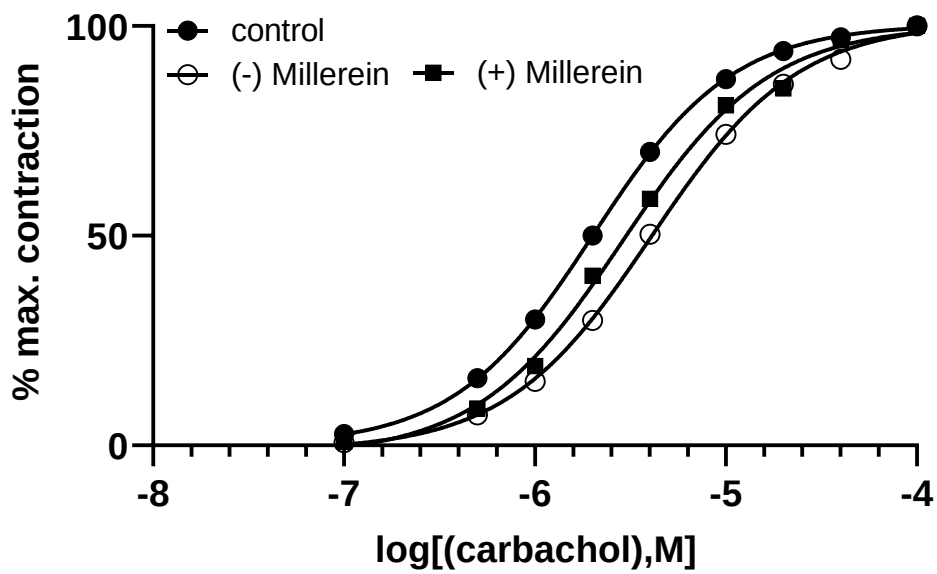

Figure 2. *Effects of millerein isomers on guinea-pig tracheal contractions triggered by graded concentrations of carbachol.*

A) What do these data show about the mechanism of action of Millerein isomers in the guinea-pig trachea (3 marks)

Next, similar experiments were carried out to determine the concentrations of carbachol required to produce the half-maximal contractions of the guinea-pig tracheal strips in presence of various other concentrations of the Millerein isomer. Dose ratios were calculated and used to generate Figure 3.

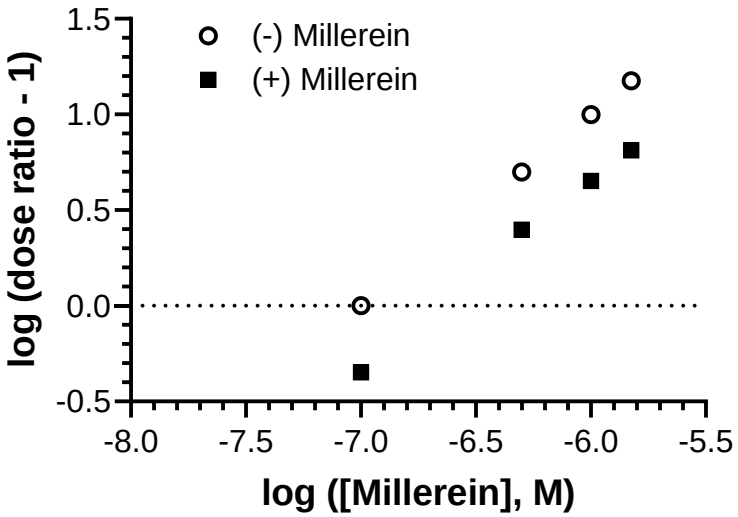

Figure 3. *Relationship of carbachol dose ratios to different concentrations of the Millerein isomers.*

B) Determine the affinity constant of each Millerein isomer for receptors mediating the contractions of guinea-pig tracheal smooth muscle. (3 marks)

In a separate experiment, the binding of [^{3}H]-N-methylscopolamine ([^{3}H]-NMS) to homogenates of the cerebral cortex of the same guinea-pigs was studied in presence and absence of 5µM of N-methylatropine (NMA). The affinity constant for [^{3}H]-NMS binding to the target receptors present in this preparation was found to be 25µM^{-1}. Afterwards, the effects of the Millerein isomers on binding of [^{3}H]-NMS to homogenate of guinea-pig cerebral cortex were studied. Plots of the concentration of each of the Millerein isomers required for 50% inhibition of the specific binding of [^{3}H]-NMS versus the concentration of [^{3}H]-NMS used were linear and had a slope of 20 and 35 for (-)-Millerein and (+)-Millerein, respectively.

C) Determine the affinity constants of the two Millerein isomers for binding to the homogenate of guinea-pig cerebellar cortex. What can you conclude about the binding of Millerein isomers in this tissue? (7 marks)

[Question 1 continues on the NEXT PAGE]

D) What concentration of the racemic mixture of (±)-Millerein (i.e. equal concentration of each isomer) would be required to occupy 50% of the target receptors present in the lungs and in the cerebral cortex of guinea-pigs? (5 marks)

E) CB2 ideally wants to develop an optically pure drug for clinical trials. Which of the Millerein isomers they should focus on for further development and why? (2 marks)

End of Question 1

[Question 2 starts on the NEXT PAGE]

Zileuton is an orally active inhibitor of 5-lipoxygenase (5-LO), used for chronic treatment of asthma. Zileuton is available in two formulations, called immediate release (IR) and controlled release (CR). Zileuton CR tablets for oral administration are triple-layer tablets, which comprise an immediate-release layer, a middle (barrier) layer, and a slow-release layer. There is no difference in the zileuton itself between the two formulations. Both formulations are supplied as tablets containing 600 mg zileuton.

In a study of the pharmacokinetics of these two formulations following oral administration, healthy volunteers were given either:

(a) one tablet (1 x 600 mg) of zileuton IR
(b) two tablets (2 x 600 mg) of zileuton IR
(c) two tablets (2 x 600 mg) of zileuton CR.

The volunteers had not eaten for at least two hours prior to administration of zileuton and continued to fast for at least six hours after. The mean plasma concentration of zileuton at different times following oral administration are shown in Table 1 and Figure 1.

Table 1. *Plasma concentrations of zileuton following oral administration of different formulations*

	Plasma concentration of zileuton (μg/ml)		
Time (hours)	1 x 600 mg Immediate release (IR)	2 x 600 mg Immediate release (IR)	2 x 600 mg Controlled release (CR)
1	4.4	8.9	2.3
2	4.3	8.5	3.3
4	2.9	5.8	4.5
6	1.9	3.8	4.5
8	1.2	2.5	3.0
10	0.8	1.6	2.1

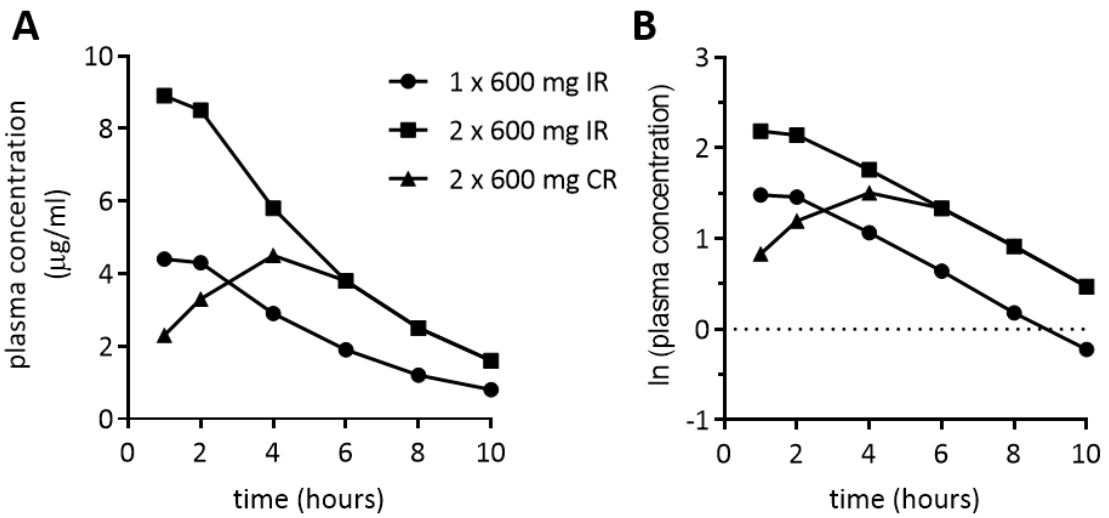

Figure 1. *Plasma concentration of zileuton (A) and natural log (ln) of plasma concentration (B).*

[Question 2 continues on the NEXT PAGE]

A) Calculate the terminal phase half-life for the two formulations of zileuton. (2 marks)

B) Calculate the clearance relative to bioavailability (CL/F) and terminal phase volume of distribution relative to bioavailability (V_d/F) of the two formulations of zileuton. (4 marks)

C) Explain why the time course of the zileuton plasma concentration is different between the three groups of volunteers. (3 marks)

In a follow-up study, the volunteers were orally administered one tablet of zileuton IR or two tablets of zileuton CR formulation. This time, however, the dose was given at the same time as a main meal containing fats and protein. The AUC for the two formulations was calculated and compared to the AUC for the same formulation and dose in fasting volunteers (calculated in the first study). The ratio between the AUCs is given in Table 2.

Table 2. *Effect of taking zileuton with a meal ('fed') on AUC*

(Dose) formulation	AUC_{fed}:$AUC_{fasting}$
(1 x 600 mg) IR	1.02
(2 x 600 mg) CR	1.36

D) Suggest why taking zileuton CR with a meal increases its AUC. (1 mark)

The recommended dosage of the controlled release formulation is 2 tablets (2 x 600 mg), twice a day.

E) Calculate the average plasma concentration of zileuton that will be achieved at steady state with this dosing schedule if the doses are (i) taken with a meal, or (ii) taken when fasting. (2 marks)

If the immediate release formulation is used, the recommended dosage is one tablet (600 mg).

F) How often should one tablet of the immediate release formulation be administered to achieve the same average plasma concentration of zileuton as the dosing schedule in (E) under fed and fasting conditions? (2 marks)

G) Explain the potential clinical benefits of using the controlled release formulation. (2 marks)

[Question 2 continues on the NEXT PAGE]

Zileuton is eliminated by metabolism *in vivo*. The major route of metabolism in humans is glucuronidation. To improve the duration of action of zileuton a series of structurally-related analogues of zileuton were synthesised. One analogue (761) was found to have a plasma half-life of 16 hours in humans.

H) What experiments could be performed to investigate why analogue 761 has a much longer plasma half-life? Clearly indicate the limitations of your experiments. (4 marks)

End of Question 2

[Question 3 starts on the NEXT PAGE]

Scientists at a leading pharmaceutical company have characterised a new drug, Rahzine. It acts on ionotropic glutamate receptors (iGluR) and is an interesting candidate for the treatment of a neurological disorder. The pharmacology of Rahzine was evaluated using whole cell patch clamp recording to measure the current across the membrane of an HEK293 cell, which expressed glutamate receptors. Recordings were made from different cells expressing different glutamate receptor isoforms (dubbed here W, X, Y, Z). The extracellular solution contains 140 mM Na^{2+} as the principle carrier ion. Drug(s) were applied to the cell during the recording and any effect on opening or closing of the glutamate receptor, and so on the current, was detected and measured. See **Figure 1**.

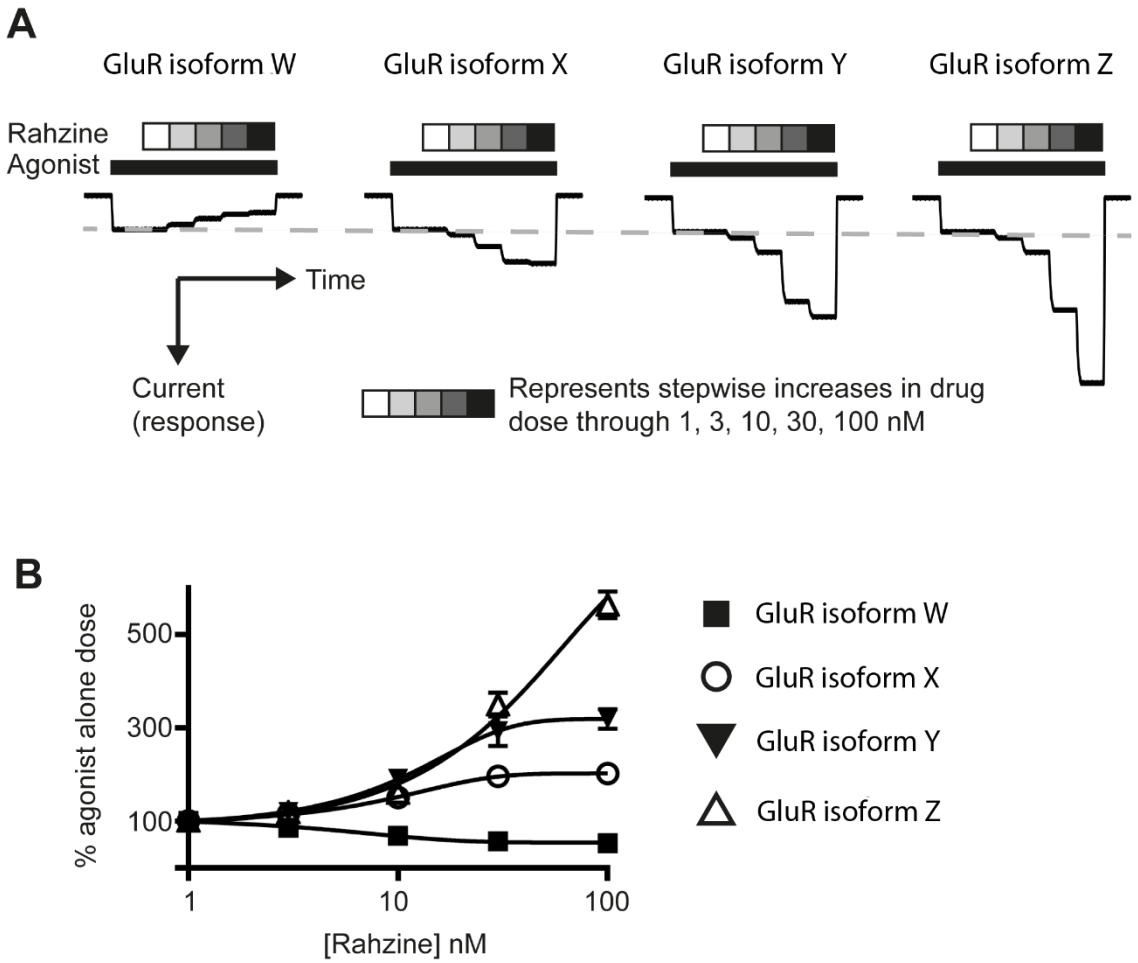

Figure 1. A. *Raw data recordings from cells expressing glutamate receptor isoforms W, X, Y or Z. An increase in sodium current through the receptor channels is observed as a downward response. Black bar above recordings represents the period during which a saturating dose of the agonist was applied. In addition to the saturating dose of agonist, Rahzine was subsequently applied in incrementally increasing concentrations (graded bar).* ***B.*** *Dose response curves of the effect of Rahzine on responses, plotted from the recordings, mean ± standard error, for n = 4 different cells.*

[Question 3 continues on the NEXT PAGE]

A) What conclusions can be drawn from Figure 1 about Rahzine? (3 marks)

B) Explain why it is <u>not</u> possible to know from the data in Figure 1 whether Rahzine is a modulator or an agonist. Give a very simple experiment using the same technique that would distinguish between the two actions. (2 marks)

A derivative of Rahzine named Rahzinide was later developed and tested in the same assay in combination with Rahzine. See **Figure 2**.

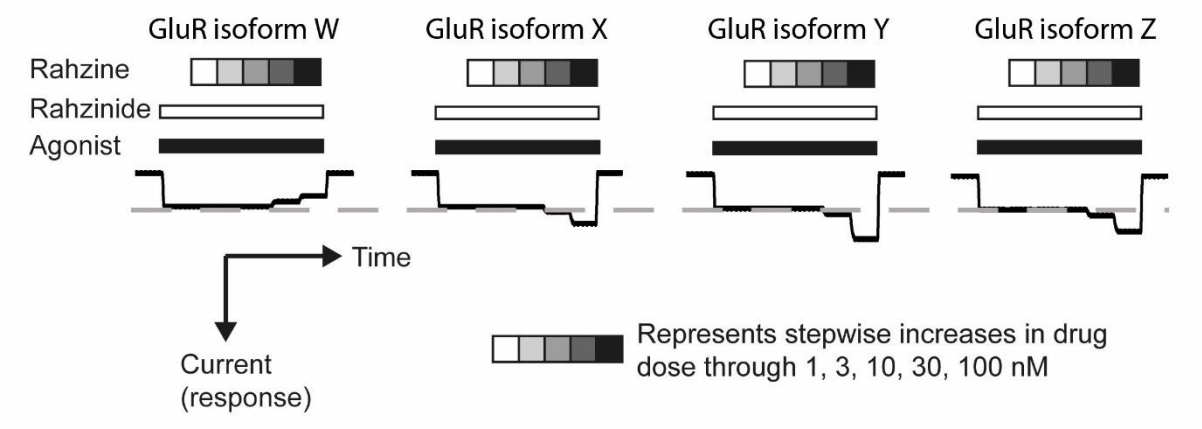

Figure 2. *Raw data recordings from cells expressing different glutamate receptor isoforms W, X, Y or Z. The black bar above the recording represents the period during which a saturating dose of agonist was applied, and the white bar represents a dose of Rahzinide. Rahzine was subsequently applied in incrementally increasing concentrations (graded bar).*

C) What conclusions can be drawn about Rahzinide? Suggest an experiment (does not have to be the same technique) to learn more about the mechanism of action of Rahzinide. (2 marks)

The scientists then decided to further investigate the action of the original compound, Rahzine. To do this they used an alternative related technique called single channel recording. Instead of observing a whole cell population of receptors opening and closing in response to drug (as was done in Figure 1 & 2), this technique observes only one single channel opening and closing. As before, a saturating concentration of agonist was used. This was followed by subsequent addition of a saturating Rahzine dose. The glutamate receptor channel transitions stepwise between being closed or open. The size of the 'step' represents the speed with which Na^+ ions flow through the channel, known as the conductance. See **Figure 3**.

[Question 3 continues on the NEXT PAGE]

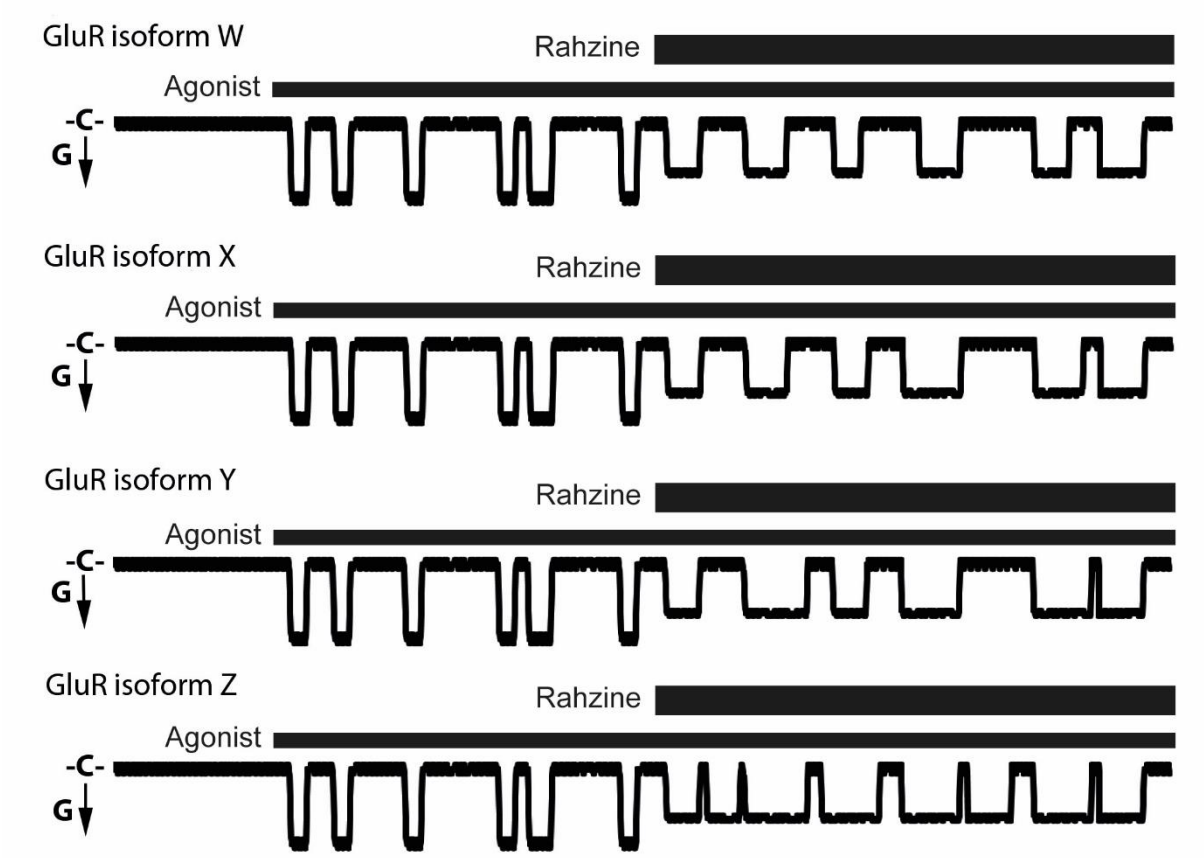

Figure 3. *Four single channel recordings from glutamate receptor isoforms W, X, Y or Z. Thin black bar above recordings represents the period during which a saturating concentration of agonist was applied. In addition to the saturating concentration of agonist, a saturating concentration of Rahzine was subsequently applied, indicated by thick black bar. The level at -C- marks the closed state. The vertical axis (G) measures the conductance of the open channel state.*

D) What conclusions can be drawn about Rahzine from the single channel recordings alone? (2 marks)

E) Explain why the data in Figure 3 suggests Rahzine is a modulator rather than an agonist. (2 marks)

F) Explain the results from Figure 1 in light of the additional data in figure 3 (2 marks)

Next the scientists generated a second derivative of Rahzine, called Rahzinol. The same whole cell patch clamp experiments were performed as originally done in Figure 1, but using Rahzinol instead. The dose response curves are plotted in **Figure 4**.

[Question 3 continues on the NEXT PAGE]

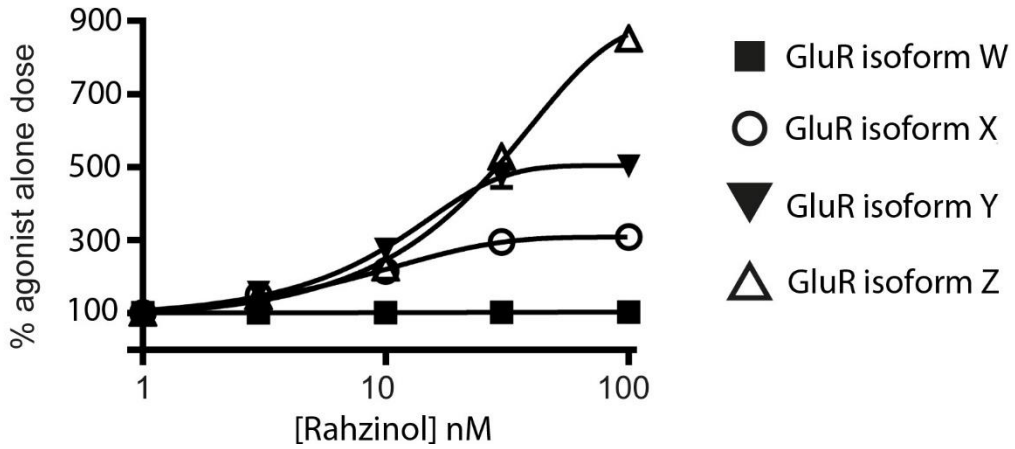

Figure 4. *Dose response curves of Rahzinol modulation of agonist responses, mean ± standard error, for n = 4 different cells.*

G) How might the mechanism of action of Rahzinol differ from Rahzine to explain these results. (2 marks)

In the first four experiments (Figures 1-4) the recordings were made using an extracellular solution containing a high concentration of Na^+ (140 mM), intended to be the principle carrier ion, and low concentration of Ca^{2+} (2 mm). Next, the experiment was repeated but all of the Na^+ was replaced with Ca^{2+}, intended to make Ca^{2+} the principle carrier ion. See **Figure 5**.

[Question 3 continues on the NEXT PAGE]

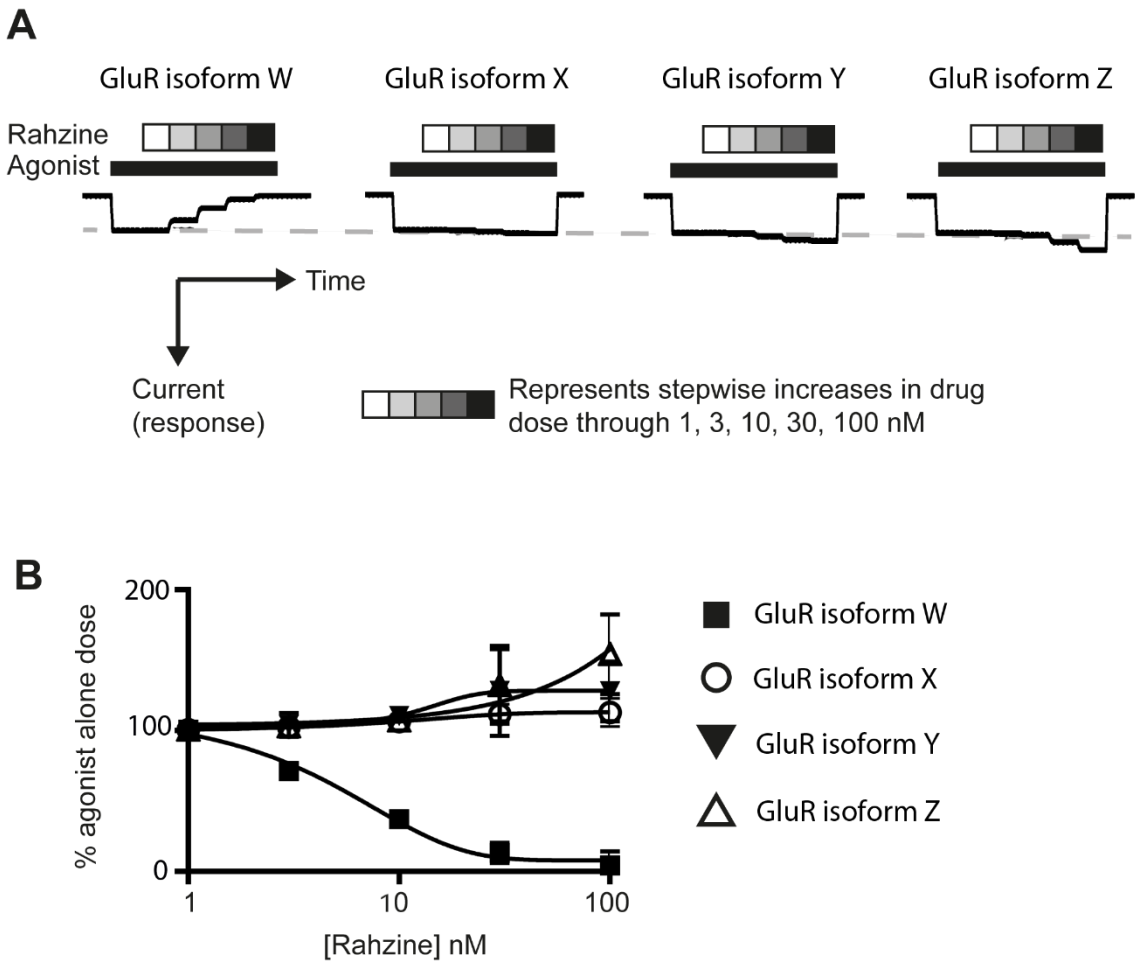

Figure 5. A. *Raw data recordings from cells expressing different glutamate receptor isoforms W, X, Y or Z). An increase in Ca^{2+} current through the receptor channels is observed as a downward response. Black bar above recordings represents the period during which a saturating concentration of agonist was applied. In addition to the saturating concentration of agonist, Rahzine was subsequently applied in incrementally increasing concentrations (graded bar).* ***B.*** *Dose response curves of Rahzine modulation of agonist responses, plotted from the recordings, mean ± standard error, for n = 4 different cells.*

H) What effect does using Ca^{2+} have on current for the agonist responses and for the Rahzine modulation of the agonist responses? (2 marks)

I) What could be the possible cause of the different effect of Rahzine on Ca^{2+} current versus Na^{+} current. (1 mark)

J) Which class of ionotropic glutamate receptors are these subtypes most likely to belong to? Explain why. (1 mark)

K) Can you envisage a situation in which Rahzine might be therapeutically useful and explain why? (1 mark)

END OF PAPER